MATH 202 Your name: _____

Learning target T1, version 1

Find the Taylor polynomials of a given function about a given center a.

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1. Suppose that $f(x)$ is some function, and that we know the following information:

$$f(-1) = 3,\ f'(-1) = \frac{1}{2},\ f''(-1) = -1,\ f'''(-1) = 0,\ f^{(4)}(-1) = \frac{1}{5}$$

   Determine $P_4(x)$, the degree 4 Taylor polynomial approximation to $y = f(x)$ centered at $a = -1$.

2. Now suppose that $g(x) = \dfrac{1}{x} = x^{-1}$. Find the degree 2 Taylor polynomial approximation, $P_2(x)$, to $g(x)$, centered at $a = 3$. Show your work!

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Learning target T1, version 2

Find the Taylor polynomials of a given function about a given center a.

Evaluate each of the following integrals. For each one:

- show all your work, including any new variables you use,
- and write one sentence that explains your choice of method and substitution.

Your process is even more important than your answer in each. Use the back of this paper if you need more room to explain your thinking.

1. $\int 5x\sin(2x)\,dx$

2. $\int e^x\cos(e^x)\,dx$

3. $\int x^7\ln(x)\,dx$

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Learning target T1, version 3

Find the Taylor polynomials of a given function about a given center a.

Evaluate each of the following integrals. For each one:

- show all your work, including any new variables you use,
- and write one sentence that explains your choice of method and substitution.

Your process is even more important than your answer in each. Use the back of this paper if you need more room to explain your thinking.

(a) $\int x^5 \cos(x^6+3)\,dx$

(b) $\int \theta^2 \sin(3\theta)\,d\theta$

(c) $\int \dfrac{7x^2}{\sqrt{4x^3+11.2534}}\,dx$

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Learning target T1, version 4

Find the Taylor polynomials of a given function about a given center a.

Evaluate each of the following integrals. For each one:

- show all your work, including any new variables you use,
- and write one sentence that explains your choice of method and substitution.

Your process is even more important than your answer in each. Use the back of this paper if you need more room to explain your thinking.

(a) $\int x \ln(x)\, dx$

(b) $\int y^5 \sin(3y^6 - 2)\, dy$

(c) $\int 5q e^{-5q}\, dq$